PLATFORM ARCHITECTURE

ZYVOR PLATFORM ARCHITECTURE

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Scope: End-to-end architecture for the full Zyvor / HyperSDK product portfolio
Last updated: June 2026

Table of Contents

1. Executive Summary
2. Platform Vision & Layer Model
3. Install Order & Lifecycle
4. Foundation Layer
5. Cloud Control Layer
6. Application Layer
7. Migration & Portability Layer
8. Observability & Intelligence Layer
9. Marketing & Reference
10. Development Umbrella & Utilities
11. Upstream & Reference Projects
12. Cross-Project Integration Matrix
13. Technology & Port Reference
14. Security Boundaries
15. Deployment Patterns
16. Per-Project Documentation Index

1. Executive Summary

The **Zyvor platform** is a vertically integrated stack for running enterprise infrastructure ? from bare metal through Kubernetes, KubeVirt virtual machines, AI/GPU workloads, and VM migration from legacy hypervisors. Products are deliberately **modular siblings**, not a monolith: each repo owns a domain, exposes stable APIs, and composes with others through kubeconfig handoffs, REST integrations, and shared migration pipelines.

```
????????????????????????????????????????????????????????????????????????????????????????
?                        ZYVOR PLATFORM ? AT A GLANCE                                  ?
????????????????????????????????????????????????????????????????????????????????????????
?  Day 0        hypercluster          Bare-metal K8s bootstrap (Kubespray/k3s)?
?  Physical     machina · zyvor-fabric · IronWolf   Hypervisor & metal layers ?
?  Cloud        v9s · VMRogue · ragnarok            KubeVirt command centers  ?
?  AI           forge                               GPU/LLM/FinOps control    ?
?  Apps         hermes                              K8s application gateway   ?
?  Migration    hypersdk · hyper2kvm · guestkit     Export ? inspect ? deploy ?
?  Portability  Aether                              Multi-runtime workloads   ?
?  Network      packetwolf                          Cilium/eBPF intelligence  ?
?  Marketing    hypersdk-web                        zyvor.dev product site    ?
```

??
Design philosophy:
Principle · Implementation
Composable, not monolithic · hypercluster bootstraps; Forge/Hermes/Zeus install *into* the cluster
Same problem, one owner · VM migration = hyper2kvm; disk inspect = guestkit; network intel = packetwolf
Kubernetes-native where possible · Operators, CRDs, Helm charts; systemd for bare-metal daemons
Offline-first migration · GuestKit + hyper2kvm fix guests *before* boot on KVM
AI as embedded ops · Forge Zeus, Ragnarok healer, Veyron copilot ? not bolt-on chatbots
Commercial umbrella: zyvor.dev · GitHub org: ssahani

2. Platform Vision & Layer Model

The platform follows a **bottom-up dependency stack**. Lower layers provide compute, storage, and networking; upper layers add orchestration, UX, and intelligence.

```
flowchart TB
 subgraph L0["Layer 0 ? Physical & Metal"]
  MACH[machina<br/>libvirt/KVM]
  FAB[zyvor-fabric<br/>systemd-vmspawn]
  IW[IronWolf<br/>Metal3/BMC]
 end

 subgraph L1["Layer 1 ? Cluster Bootstrap"]
  HC[hypercluster<br/>Kubespray / k3s]
 end

 subgraph L2["Layer 2 ? Cloud Control"]
  ZEUS[v9s / Zeus OS]
  VYR[VMRogue / Veyron]
  RAG[ragnarok]
 end

 subgraph L3["Layer 3 ? AI Infrastructure"]
  FORGE[forge]
 end

 subgraph L4["Layer 4 ? Application Experience"]
  HERM[hermes]
 end

 subgraph L5["Layer 5 ? Migration & Portability"]
  HSDK[hypersdk]
  H2K[hyper2kvm]
  GK[guestkit]
  AETH[Aether]
 end

 subgraph L6["Layer 6 ? Network Intelligence"]
  PW[packetwolf]
 end
```

MACH --> HC

FAB --> HC

IW --> HC

HC --> ZEUS

HC --> VYR

HC --> RAG

HC --> FORGE

HC --> PW

ZEUS --> HERM

FORGE --> HERM

HSDK --> H2K

H2K --> GK

GK --> VYR

AETH -.-> ZEUS

PW --> FORGE

PW --> VYR

Layer Responsibilities:

Layer · Domain · Primary Products · Runtime
0 ? Physical · Hypervisor OS, bare-metal provisioning · machina, zyvor-fabric, IronWolf · Linux host, systemd
1 ? Bootstrap · K8s cluster lifecycle · hypercluster · Ansible, k3s, SSH
2 ? Cloud control · KubeVirt VM operations · v9s, VMRogue, ragnarok · In-cluster + sidecar APIs
3 ? AI infra · GPU, LLM, FinOps, Zeus SRE · forge · K8s operators + microservices
4 ? Applications · App discovery & gateway · hermes · K8s controller + Rust gateway
5 ? Migration · VM export, inspect, deploy · hypersdk, hyper2kvm, guestkit, Aether · Host daemon + CLI
6 ? Network intel · eBPF, Cilium, AutoPolicy · packetwolf · DaemonSet + host CLI

3. Install Order & Lifecycle

Reference install path for a greenfield Zyvor deployment. See also zyvor-platform-stack.md.

[Optional] Metal / Hypervisor [Optional] Migration / Inspect
? ?
? ?
machina · zyvor-fabric · IronWolf guestkit · hyper2kvm
? ?

??
?
hypercluster ??? Day 0: preflight ? deploy ? test
?
??????????????????????????????????????
? ? ?
packetwolf forge v9s (Zeus OS)
(needs Cilium) (GPU / AI) (KubeVirt ops)
? ? ?
??????????????????????????????????????
?
ragnarok · VMRogue · Aether · hermes (as needed)

Phase Reference:

Phase · Product · Trigger · Output
0a · machina / zyvor-fabric · Need libvirt or systemd-native VMs on metal · Running hypervisor daemon
0b · IronWolf · Need Metal3 bare-metal fleet · Provisioned BareMetalHosts
0c · guestkit + hyper2kvm · Migrating from VMware/Azure/Hyper-V · Converted QCOW2 + fix plans
1 · hypercluster · Always first for K8s · KUBECONFIG, KubeVirt/CDI optional
2 · packetwolf · Cilium-enabled cluster · Network intelligence plane
3a · forge · GPU/AI workloads · 60+ CRDs, Zeus, operators
3b · v9s / VMRogue / ragnarok · KubeVirt VM UX · VM command centers
4 · hermes · Application layer UX · /a/{ns}/{app} gateway
5 · Aether · Multi-runtime portability · Unified deploy/migrate

Forge Handoff (Day 2):

After hypercluster deploy:

```
./hypercluster -c cluster.conf forge-handoff --json > handoff.json
```

 # From forge checkout:

```
./scripts/lib/hypercluster-handoff.sh --handoff handoff.json
```

4. Foundation Layer

4.1 hypercluster:

Repo: ssahani/hypercluster · **Role:** Day 0 Kubernetes bootstrap
Attribute · Detail
Stack · Bash orchestrator, Ansible (Kubespray), k3s shell libs, Electron desktop
Entry point · hypercluster CLI script
Key paths · hypercluster, lib/, config/cluster.conf.example, desktop/
Commands · init, preflight, deploy, scale, upgrade, destroy, kubeconfig, test, health, forge-handoff
Architecture:

cluster.conf ? hypercluster CLI ? SSH ? Ansible/k3s ? nodes
? fetch-kubeconfig ? ~/.kube/config
? optional: KubeVirt + CDI + Cilium + MetalLB

k3s path bundles: Cilium, cert-manager, MetalLB, KubeVirt, CDI. **Kubespray path:** Full HA control plane for

production.
Integrations: Bootstraps clusters consumed by Forge, Hermes, Zeus OS, Veyron, PacketWolf, Ragnarok.

4.2 machina:

Repo: ssahani/machina · **Role:** Enterprise libvirt/KVM hypervisor management
Attribute · Detail
Stack · Rust workspace (12 crates), React 19 web UI, ratatui TUI
Daemon · machina-daemon ? HTTPS :5092
Entry points · daemon/src/main.rs, machinactl, web/, tui/
Hypervisor · libvirt ? QEMU/KVM
Core components:
Component · Path · Responsibility
REST + WebSocket API · daemon/ · VM lifecycle, console proxies, RBAC
Web dashboard · web/ · 40+ pages, noVNC/xterm.js
TUI client · tui/ · Terminal VM management
Fleet controller · controller/ · Multi-host orchestration
Per-host agent · agent/ · libvirt RPC on worker nodes
Core libvirt layer · core/ · Bindings, validation, types
Data flow:

```
Web/TUI/API ? machina-daemon ? libvirt API ? QEMU/KVM
WebSocket   ? VNC/SPICE/serial/SSH proxy ? guest console
Prometheus  ? /api/v1/prometheus
```

Integrations: Physical layer under Zeus OS; KubeVirt migration path; OpenStack Nova/Glance; PacketWolf observability; hyper2kvm import target.
Active dev branch: machina-hw-pr ? KubeVirt hardware drawers, ConsoleHub cinema mode, PCI/USB passthrough API.

4.3 zyvor-fabric:

Repo: ssahani/zyvor-fabric · **Role:** systemd-native private cloud control plane
Attribute · Detail
Stack · Rust (40 crates), React web UI, K8s operator, Terraform provider
Daemon · vmspawnd ? HTTPS :8443, 480+ REST endpoints
Runtime · systemd-vmspawn, systemd-machined, KVM
CLI · vmctl, vmctl-tui
Architecture:

```
vmctl / TUI / Web / TF / Operator ? vmspawnd ? vmspawn-driver ? systemd-vmspawn ? KVM
                ? D-Bus ? machined ? scope management
```

Key paths: backend/vmspawnd/, backend/vmspawn-driver/, backend/crates/machinectl-driver/, backend/networking/, backend/storage/, backend/gpu-passthrough/, backend/ha/, integrations/machina/
Positioning: Alternative to machina for operators who prefer systemd-native VM management over libvirt. Integrates with machina via integrations/machina/.

4.4 IronWolf:

Repo: ssahani/IronWolf · **Role:** Metal3 bare-metal automation extension
Attribute · Detail

Stack · Go controller-runtime, React dashboard, 24 CRDs (bmo.metal3.io)
Dashboard · :8888
Extends · Metal3 BareMetal Operator
Core CRDs & controllers:
Resource · Purpose
ClusterClaim · Hardware allocation (binpack/spread/topology)
FirmwareProfile · Fleet firmware lifecycle
HardwareBackup · BMC configuration backup
RemediationStrategy · Progressive auto-escalation
BMC Reconciler · IPMI/Redfish/iLO/iDRAC
Data flow:

ClusterClaim ? scheduler ? BareMetalHost ? Ironic provision ? K8s node
FirmwareProfile ? BMC Reconciler ? firmware update ? status CR
Dashboard ? WebSocket ? controller events

Integrations: Feeds bare metal into hypercluster and Aether Metal3 runtime.

5. Cloud Control Layer

Three products address KubeVirt VM management with different emphases. They overlap in domain but target different personas and feature depth.
Product · GitHub · Emphasis · Default Port
v9s (Zeus OS) · zeus-os · Visual OS for K8s/VMs; 48+ TUI views; enterprise auth · :5050
VMRogue (Veyron) · Veyron · GitOps CRDs, templates, Mission Control · :8080
ragnarok · ragnarok · AI-powered VM ops, NL provisioning, cost optimizer · Helm Ingress

5.1 v9s (Zeus OS):

Repo: ssahani/zeus-os · Local folder: v9s
Attribute · Detail
Stack · Rust workspace (core, tui, web, cli), kube-rs, React UI
Surfaces · Web dashboard, 48+ TUI views, CLI
Auth · OIDC, LDAP, SAML, tenants, API tokens, audit
Key paths: core/ (KubeVirt client), tui/, ui/, cli/, chart/, integrations/, runbooks/
Capabilities: VM lifecycle, hot-plug, live migration, VNC/serial/SSH consoles with WebSocket metrics, cost visibility, topology/RCA hooks, Velero integration.
Data flow:

zeus-os TUI/web ? kube-rs ? KubeVirt API ? VirtualMachine/VMI CRs
WebSocket watch ? live state ? dashboards/consoles
OIDC ? session ? scoped API tokens

Positioning: Primary cloud-layer command center; hosts Hermes application layer.

5.2 VMRogue (Veyron):

Repo: ssahani/Veyron · Local folder: VMRogue
Attribute · Detail
Stack · Rust core (veyron binary), Go operator, React Mission Control
CRDs · VeyronVM, VeyronPolicy (CEL deny rules)

Templates · 44 OS templates, 8 resource profiles
Key paths:
Path · Purpose
src/api/server.rs · REST API (49+ route groups)
src/kube/guestkit_client.rs · GuestKit pre-flight integration
src/templates/ · LAMP, K8s, 3-tier, CI/CD blueprints
src/gitops/ · GitOps export/sync
src/migration/ · Live/offline migration strategies
operator/main.go · VeyronVM CRD reconciler
terraform-provider-vmrogue/ · Terraform provider
Capabilities: Template/blueprint engine, declarative VeyronVM CRs, direct K8s VNC (no virtctl timeout), GuestKit assurance, PacketWolf network context, copilot handlers.

5.3 ragnarok:

Repo: ssahani/ragnarok
Attribute · Detail
Stack · Rust backend (Axum, kube-rs, SQLx/PostgreSQL), React frontend, KubeVirt operator
AI features · NL provisioning, AI healer, cost optimizer
Deploy · Helm, Terraform, Ansible, Playwright E2E
Key paths: backend/, frontend/, operator/, charts/ragnarok/, docs/AI_NL.md
Integrations: Optional composite with Aether via RAGNAROK_URL for attestation.

6. Application Layer

6.1 hermes:

Repo: ssahani/hermes · **Role:** Application Operating Layer for Kubernetes
Attribute · Detail
Stack · Go controller, Rust server (Axum), React UI, SQLite catalog
Gateway · /a/{namespace}/{slug}/* reverse proxy + WebSocket
Default port · :31847
Architecture:

K8s Services ? Go Controller ? SQLite catalog ? Rust Server ? React UI
User click ? Gateway /a/ns/app ? ClusterIP Service (HTTP/WebSocket proxy)

Key paths: cmd/hermes-controller/, crates/hermes-server/, crates/hermes-gateway/, ui/, charts/hermes/
Recognition engine: Auto-discovers Grafana, Argo CD, Harbor, Zeus/v9s, Forge via labels/annotations. Application memory: favorites, recents, spotlight search.
Positioning: Launchpad-style UX on top of Zeus OS clusters.

7. Migration & Portability Layer

7.1 End-to-End Migration Pipeline:

sequenceDiagram
participant Src as Source Hypervisor
participant HSDK as hypersdk

participant GK as guestkit
participant H2K as hyper2kvm
participant Target as machina / KubeVirt

Src->>HSDK: discover + export VMs
HSDK->>HSDK: job queue ? disk images
HSDK->>GK: disk path for assurance
GK->>GK: boot score + fix plan
GK->>H2K: migrate-plan YAML
H2K->>H2K: VMCraft convert + offline fixers
H2K->>Target: virsh define OR PVC + VM CR

7.2 hypersdk:

Repo: ssahani/hypersdk- · Public: hypersdk/hypersdk
Attribute · Detail
Stack · Go 1.25, chi router, bbolt/SQLite, embedded React dashboard
Daemon · hypervisord ? :5080
CLI binaries · hyperctl, hyperexport, + 6 more
Providers · vSphere, Nutanix, AWS, Azure, GCP, Hyper-V, + others
Key paths: cmd/hypervisord/, daemon/api/server.go (238 routes), daemon/jobs/manager.go, providers/, web/dashboard-react/, openapi.yaml
Architecture: Hexagonal provider adapters ? job scheduler ? export pipeline (discover ? export ? convert ? manifest) ? WebSocket progress to dashboard.

7.3 hyper2kvm:

Repo: ssahani/hyper2kvm- · Public: hypersdk/hyper2kvm
Attribute · Detail
Stack · Python VMCraft engine, Go h2kweb API, React dashboard, K8s operator
Web API · h2kweb ? HTTPS :5070 (110+ endpoints)
CLI · h2kvmctl, h2k, zkvm TUI (Go/Bubble Tea)
Key paths: hyper2kvm/vmcraft/, hyper2kvm/pipeline/, hyper2kvm/fixers/, hyper2kvm/providers/, web/h2kweb/, web/dashboard/, operator/
VMCraft engine: Pure-Python disk/VM manipulation ? 5?7× faster than shell pipelines. **Offline fix engine:** 35+ OS versions, Windows VirtIO injection, GRUB/fstab/initramfs fixes.
Capabilities: Full VM lifecycle, VNC console, multi-cluster kubeconfig, libvirt?KubeVirt live migration.

7.4 guestkit:

Repo: hypersdk/guestkit · Local: guestkit
Attribute · Detail
Stack · Pure Rust (no libguestfs), clap CLI, ratatui TUI (guestctl)
Bindings · PyO3 ? hypersdk-guestkit on PyPI
Disk formats · QCOW2, VMDK, RAW
Key paths: src/disk/, src/core/, crates/guestkit-agent-protocol/, crates/zyvor-guest-agent/, k8s/
Commands:
Command · Output
doctor · Boot probability + blockers

migrate-plan · Hypervisor-aware fix YAML
fleet analyze · Multi-VM assurance report
Disk intelligence: MBR/GPT, ext4/NTFS/XFS detection, Windows registry parsing ? all without booting the guest.

7.5 Aether:

Repo: ssahani/Aether · **Role:** Universal runtime portability control plane
Attribute · Detail
Stack · Rust (~37K LOC), React 18 web UI, Helm, deb/rpm
Runtimes · Podman, Kubernetes, KubeVirt, Metal3
Migration · 16 runtime-pair combinations
Default port · :8080
Key paths: src/adapters/, src/migration/, src/intelligence/, src/drift/, src/api/, web/, helm/
Core abstractions:
Component · Responsibility
Unified Workload Spec · Single YAML validated across four runtimes
Intent Engine · Scores cost, latency, reliability ? runtime selection
Migration Engine · Blue-green, rolling, immediate with rollback
Policy Gate · OPA-style checks, RBAC API
Drift Detector · Live reconciliation loop
Analogy: Terraform for *where* workloads run ? not just *what* they are.

8. Observability & Intelligence Layer

8.1 forge:

Repo: ssahani/forge · **Role:** Kubernetes-native AI infrastructure control plane
Attribute · Detail
Stack · Go operators, Python microservices (FastAPI), Rust CLI, React web-ui (94 pages), 24 eBPF programs
CRDs · 60+ custom resources
API Gateway · :24631
Service map:
Service · Port · Role
API Gateway · 24631 · Auth, tenancy, aggregation
Intelligence Engine · 8090 · Closed-loop optimizers
Zeus · 8091 · AI ops engineer, multi-LLM agents
Topology Engine · 8092 · Service topology
Incident Engine · 8093 · Incident correlation
Model Router · 8094 · LLM routing
Embedding Service · 8095 · Vector embeddings
Operators: GPU, AI, Quota, Network Intelligence, Virtualization.
Architecture layers:

```
Clients (UI/CLI/SDK) ? API Gateway ? Zeus | Intelligence | Operators
Intelligence tick ? optimizers ? remediation plan ? ActionBroker ? K8s mutations
GPU nodes ? node-agent ? DCGM ? Prometheus ? dashboards
```

Design principles: Kubernetes-native first, closed-loop intelligence, Zeus embedded (not bolted on), defense in depth, graceful degradation, air-gap readiness.

Full detail: forge/docs/ARCHITECTURE.md

8.2 packetwolf:
Repo: ssahani/packetwolf · **Role:** Kernel-native network intelligence for Kubernetes
Attribute · Detail
Stack · Rust (~71K LOC), eBPF programs, Axum web-api (130+ routes), 18-tab TUI
CLI · netpred
Requires · Cilium on cluster
Default port · :8443
Key paths: src/modules/ (15 intelligence modules), src/ebpf/, src/policies/, web-api/, netpred-ebpf/, chart/
Capabilities:
Feature · Description
God Mode explain · Cross-layer root cause: netpred explain <service>
AutoPolicy · Traffic learning ? Cilium NetworkPolicy suggestions
Autonomous Healer · Network remediation
Process attribution · Process-to-socket mapping
Integrations: Intelligence plane for Forge; Veyron PacketWolf handler; requires Cilium.

9. Marketing & Reference

9.1 hypersdk-web:
Repo: ssahani/hypersdk-web · **Role:** Enterprise marketing website (not a runtime product)
Attribute · Detail
Stack · Docusaurus 3, React 19, Go contact-mailer
Published · zyvor.dev
Content · 52 pages, 16 blog posts, 30+ PDF presentations
Architecture:

Static Docusaurus build ? nginx :443 ? contact form ? Go mailer
Links to live products: hypersdk (:5080), hyper2kvm (:5070).

10. Development Umbrella & Utilities

10.1 tt (meta-workspace):
Path: /Users/ssahani/tt/tt/ · **Role:** Local cross-project development umbrella (not a deployable product)
Subproject · Purpose
hypersdk/ · Go VM export daemon (mirror of hypersdk-)
hyper2kvm/ · Python migration toolkit (mirror of hyper2kvm-)
netctl/ · Rust CLI for systemd-networkd declarative config
netevd/ · Rust network event daemon (netlink, hooks, REST, Prometheus)
cloud-netconfig/ · Multi-cloud instance network config (AWS/Azure/GCP metadata)
hypersdk-org-profile/ · GitHub org profile assets
Supporting infrastructure for machina, zyvor-fabric, and migration VMs.

11. Upstream & Reference Projects

These projects live in the workspace for integration reference but are **not Zyvor products**:
Project · Upstream · Role in workspace
cockpit · cockpit-project/cockpit · Linux server admin UI reference
cockpit-machines · cockpit-project/cockpit-machines · libvirt VM module reference
machina-hw-pr · ssahani/machina (dev branch) · Active KubeVirt hardware integration PR

12. Cross-Project Integration Matrix

From · To · Integration Mechanism
hypercluster · Forge, Hermes, Zeus, Veyron · kubeconfig handoff, forge-handoff --json
machina · v9s · Physical hypervisor ? cloud VM layer
machina · hyper2kvm · VM import/migration to libvirt
hypersdk · hyper2kvm · Export ? convert/deploy pipeline
guestkit · hyper2kvm, Veyron · Pre-flight boot scoring + fix plans
guestkit · Veyron · src/kube/guestkit_client.rs
Aether · All runtimes · Unified deploy/migrate across Podman/K8s/KubeVirt/Metal3
packetwolf · forge · Intelligence plane network module
packetwolf · Veyron · src/api/handlers/packetwolf.rs
hermes · v9s · Application layer on Zeus OS clusters
IronWolf · hypercluster, Aether · Bare-metal provisioning ? K8s/Metal3
forge · packetwolf, Zeus · AI ops + network intelligence
zyvor-fabric · machina · Cross-product via integrations/machina/
Aether · ragnarok · Optional attestation via RAGNAROK_URL

13. Technology & Port Reference

Language Distribution:

Language · Products
Rust · Aether, machina, zyvor-fabric, v9s, VMRogue, ragnarok, packetwolf, guestkit, forge CLI, hermes server, netctl, netevd
Go · forge operators, hypersdk, IronWolf, Veyron operator, hermes controller, hyper2kvm h2kweb
Python · hyper2kvm VMCraft, forge services
TypeScript/React · All web UIs, hypersdk-web
Bash/Ansible · hypercluster
C · cockpit (upstream)

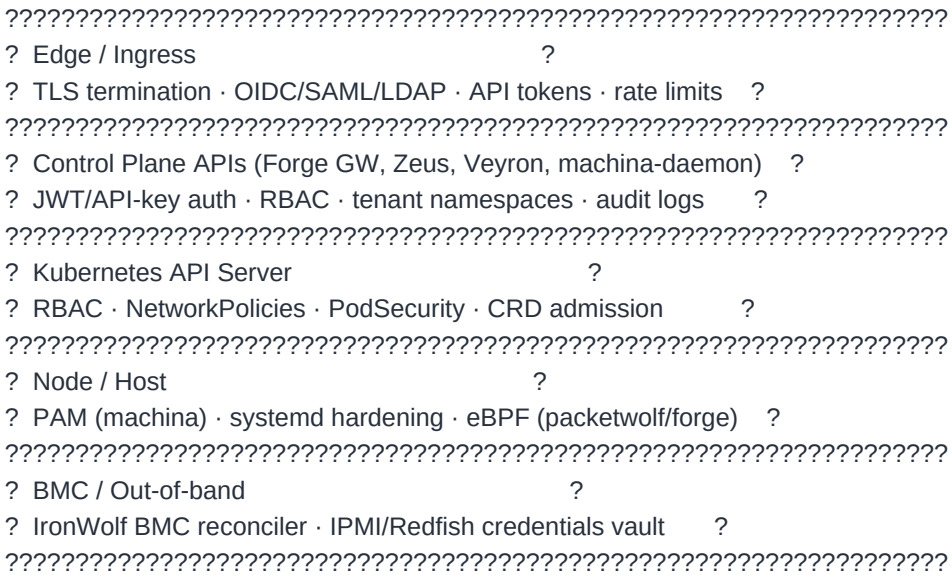
Default Service Ports:

Product · Port · Protocol
v9s (Zeus OS) web · 5050 · HTTP
hyper2kvm h2kweb · 5070 · HTTPS
hypersdk dashboard · 5080 · HTTPS
machina daemon · 5092 · HTTPS
Veyron (VMRogue) · 8080 · HTTPS
Aether API · 8080 · HTTP
Forge API Gateway · 24631 · HTTP
Forge Zeus · 8091 · HTTP

Forge Intelligence · 8090 · HTTP
Hermes · 31847 · HTTP
zyvor-fabric / packetwolf web · 8443 · HTTPS
IronWolf dashboard · 8888 · HTTP

CRD Inventory (Summary):
Product · CRD Count · API Group
forge · 60+ · forge.*
IronWolf · 24 · bmo.metal3.io
Veyron · 2+ · VeyronVM, VeyronPolicy
packetwolf · 5 · netpred.*
ragnarok · operator-managed · KubeVirt extensions

14. Security Boundaries



Key patterns:
- **Forge:** API governance middleware, per-tenant rate limits, FabricAutoPolicy gates, optional Vault CSI
- **machina / zyvor-fabric:** PAM auth, RBAC, encryption at rest, audit trails
- **Aether:** OPA-style policy gate, RBAC API (RbacStore)
- **Veyron:** VeyronPolicy CEL deny rules
- **Migration tools:** Credential store in hypervisord (bbolt); offline disk analysis (no network exposure)

15. Deployment Patterns

Pattern · Products · Install Method
Single binary + systemd · Aether, machina, guestkit, hypersdk, zyvor-fabric, netevd, netctl · cargo install / make install + systemd unit
K8s Helm operator · Forge, Hermes, IronWolf, Veyron, Ragnarok, PacketWolf, hyper2kvm · helm upgrade --install
Bash/Ansible bootstrap · hypercluster · ./hypercluster deploy from operator laptop
Python pip + Go daemon · hyper2kvm · pip install hyper2kvm + h2kweb systemd
Static marketing site · hypersdk-web · npm run build ? nginx

Upstream distro packages · cockpit, cockpit-machines · dnf/apt packages
Meta-workspace · tt · Per-subproject build; no unified deploy

Recommended Production Stack:

IronWolf (metal) ? hypercluster (Kubespray HA) ? Cilium + KubeVirt/CDI
? packetwolf ? forge (GPU) ? v9s ? hermes
? hypersdk + guestkit + hyper2kvm (migration lane)
? Aether (portability overlay, optional)

16. Per-Project Documentation Index

Project · Canonical Architecture Doc
Aether · Aether/docs/architecture/ARCHITECTURE.md
forge · forge/docs/ARCHITECTURE.md
guestkit · guestkit/README.md
hermes · hermes/docs/architecture.md
hyper2kvm · hyper2kvm-/docs/reference/architecture.md
hypercluster · hypercluster/docs/guides/zyvor-platform-stack.md
hypersdk · hypersdk-/docs/architecture/ARCHITECTURE.md
IronWolf · IronWolf/docs/architecture.md
machina · machina/README.md, machina/docs/
packetwolf · packetwolf/docs/architecture/ARCHITECTURE.md
ragnarok · ragnarok/docs/ARCHITECTURE.md
v9s · v9s/docs/ZEUS_OS.md
VMRogue · VMRogue/README.md
zyvor-fabric · zyvor-fabric/ARCHITECTURE.md
Platform (this doc) · hypercluster/docs/PLATFORM_ARCHITECTURE.md
Legacy deep dives: docs/FULL_README_LEGACY.md in Aether, hermes, hyper2kvm-, hypercluster, hypersdk-, IronWolf, machina, packetwolf, ragnarok, v9s, VMRogue.

Appendix A ? Domain Ownership (Non-Goals)

Products must **not** re-implement sibling domains. Reference: hypercluster agent training § Scope.
Domain · Owner · Do NOT build in
vLLM / Triton / KServe · forge · hypercluster
VM vCenter-style cockpit · v9s, ragnarok, VMRogue · hypercluster
Bare-metal hypervisor UI · machina, zyvor-fabric · hypercluster
VMware ? KVM migration · hyper2kvm + guestkit · hypercluster
Network auto-policy / eBPF · packetwolf · hypercluster
Full observability stack · forge, VMRogue · hypercluster
GitOps / Tekton · forge/gitops · hypercluster
Multi-runtime portability · Aether · single-cluster products

Appendix B ? GitHub Repository Map

Local Folder · GitHub Repository · Visibility
Aether · ssahani/Aether · Private

forge · ssahani/forge · Private
guestkit · hypersdk/guestkit · Public
hermes · ssahani/hermes · Private
hyper2kvm- · ssahani/hyper2kvm- · Private
hypercluster · ssahani/hypercluster · Private
hypersdk- · ssahani/hypersdk- · Private
hypersdk-web · ssahani/hypersdk-web · Private
IronWolf · ssahani/IronWolf · Private
machina · ssahani/machina · Private
packetwolf · ssahani/packetwolf · Private
ragnarok · ssahani/ragnarok · Private
v9s · ssahani/zeus-os · Private
VMRogue · ssahani/Veyron · Private
zyvor-fabric · ssahani/zyvor-fabric · Private

Document version: 2026-06-17 · Maintainer: ssahani · Platform codename: Zyvor